The impact of punctuated infectiousness on infectious disease dynamics

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Abstract

1 Introduction

The trajectory of an epidemic depends on two fundamental quantities: the basic reproduction number (R_0), which is the expected number of secondary infections that an infectious person will produce in an otherwise susceptible population; and the generation interval distribution ($g(\tau)$), which is the distribution of times at which those secondary infections are expected to occur relative to the index case's infection time ($\tau = 0$) [Wallinga and Lipsitch, 2007]. These quantities can be combined into a single object, the "infectiousness profile", defined as $A(\tau) = R_0 g(\tau)$, which describes a person's expected contribution to transmission and how it is distributed over time [Breda et al., 2012].

The reproduction number and the generation interval distribution are both expectations taken over a large population. At the individual level, the individual reproduction number, ν_i, is the expected number of infections that a particular person i will produce, such that $R_0 = E_i[\nu_i]$ [Lloyd-Smith et al., 2005]. Similarly, the "individual infectiousness profile" or "infectiousness kernel", $a_i(\tau)$, describes how a particular person's infectiousness is distributed over time, with $A(\tau) = E_i[a_i(\tau)]$ [Champredon and Dushoff, 2015, Svensson, 2007].

Estimating R_0 and $g(\tau)$ (and thus $A(\tau)$) is critical for modeling epidemic dynamics and designing interventions. However, the shape of the individual infectiousness profile $a_i(\tau)$ is poorly understood for many pathogens. There are two main reasons for this: we lack a clear understanding of whether, and how, the shape of $a_i(\tau)$ impacts epidemic dynamics beyond what $A(\tau)$ can provide, and thus lack motivation to target $a_i(\tau)$ in study designs; and we lack study designs and inferential methods capable of inferring $a_i(\tau)$ independently of $A(\tau)$.

Often, a modeling choice is made to assume $a_i(\tau) \equiv A(\tau)$. However, it is likely that the full variability in $A(\tau)$ is made up of both within-individual and between-individual variation, implying $a_i(\tau)$ is generally narrower than $A(\tau)$. Emphasizing this point, some evidence suggests that the period when SARS-CoV-2 and influenza can transmit explosively is short, lasting less than a day [Goyal et al., 2021], suggesting that $a_i(\tau)$ may in some cases be much narrower than $A(\tau)$, with much of the variation in $A(\tau)$ arising from variation between individuals and on when their peak infectiousness occurs. These considerations are sometimes built into disease transmission models, but we lack a comprehensive understanding of how the punctuatedness of $a_i(\tau)$, independent of other factors, impacts disease transmission dynamics, and we lack a clear framework for incorporating $a_i(\tau)$ into the foundational disease transmission modeling frameworks upon which virtually all others are based.

At the individual level, communicating about the likely timing of a person's infectiousness is important: it is good to know if you will be very infectious for a short time, or mildly infectious for a long time, so that you can shape your behavior accordingly. Additionally, a punctuated individual infectiousness kernel could impact epidemic dynamics. Even if the epidemic has the same mean-field behavior, it could impact the stochastic dynamics via highly synchronized transmission events, especially when an outbreak is small. It could interplay with control measures, especially those designed to intervene on infectiousness at the individual level like test- or symptom-based screening and isolation. Indeed, the impact of the punctuatedness of the individual infectiousness kernel has been examined as a sensitivity analysis in various epidemiological studies [Larremore et al., 2021] **Others?**. Stochastic effects are known to cascade through macroscopic features of an epidemic [Morris et al., 2024]. However, we lack a comprehensive understanding of how the sharpness of the individual infectiousness kernel, independent of other features like the individual reproduction number, basic reproduction number, and generation interval distribution, impacts epidemic dynamics under both uncontrolled and controlled settings. Furthermore, we lack an understand-

ing of how to measure the shape of the individual infectiousness kernel reliably, along with the settings when the kernel's shape is statistically identifiable.

The main goal: holding $A(\tau)$ constant, how do different choices of $a_i(\tau)$ impact epidemic dynamics? And how would we know what $a(\tau)$ looks like?

2 The individual infectiousness profile

2.1 Expressing the Hudson–Kermack–McKendrick model in terms of individual infectiousness

Early work by Hudson, and later by Kermack and McKendrick [Kermack and McKendrick, 1927], modeled the transmission of a permanently immunizing infection using an integral equation. In modern notation [Breda et al., 2012], the force of infection is

$$F(t) = \int_0^\infty F(t-\tau)S(t-\tau)A(\tau)d\tau \tag{1}$$

where $F(t)$ is the force of infection at time t, $S(t)$ is the density of susceptible individuals in the population at time t, and $A(\tau)$ is the infectiousness profile describing the expected contribution to the force of infection from an individual who was infected τ time units ago. The incidence of disease is the product of the force of infection and the susceptible density (*i.e.*, $-\frac{dS}{dt} = F(t)S(t)$); thus, the infectiousness profile $A(\tau)$ is the fundamental object that determines how an epidemic unfolds.

The infectiousness profile is related to the basic reproduction number R_0 and the generation interval distribution $g(\tau)$:

$$R_0 = \int_0^\infty A(\tau)d\tau \tag{2}$$

$$g(\tau) = \frac{A(\tau)}{R_0} \tag{3}$$

Since $A(\tau)$ is an expectation over individuals, it does not necessarily reflect any single person's infectiousness. For example: the SEIR model is conceptually motivated by considering an individual who undergoes a latent period of length $x_i \sim \text{Exp}(\gamma_{\text{SEIR}})$ during which they are not infectious, followed by an infectious period of length $y_i \sim \text{Exp}(\alpha_{\text{SEIR}})$ during which they are infectious at level β_{SEIR}. Mathematically, this "stepwise" individual infectiousness profile is expressed as

$$a_i(\tau) = \begin{cases} \beta_{\text{SEIR}} & \tau \in [x_i, x_i + y_i] \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

This yields the population-level infectiousness kernel

$$A(\tau) = E_i[a_i(\tau)] = \lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} a_i(\tau) = \beta_{\text{SEIR}} \frac{\gamma_{\text{SEIR}}}{\gamma_{\text{SEIR}} - \alpha_{\text{SEIR}}}(e^{-\alpha_{\text{SEIR}}\tau} - e^{-\gamma_{\text{SEIR}}\tau}) \tag{5}$$

i.e., a smooth, bell-shaped curve that arises from a sum of discontinuous, stepwise $a_i(\tau)$ (**Figure XX**). Other choices of $a_i(\tau)$ can yield the same $A(\tau)$ in Eq. 5, including

$$a_i(\tau) = \beta_{\text{SEIR}} \frac{\gamma_{\text{SEIR}}}{\gamma_{\text{SEIR}} - \alpha_{\text{SEIR}}}(e^{-\alpha_{\text{SEIR}}\tau} - e^{-\gamma_{\text{SEIR}}\tau}) \tag{6}$$

i.e., a "smooth" individual infectiousness profile where each person's infectiousness is identical to the population-level average, and

$$a_i(\tau) = \frac{\beta_{\text{SEIR}}}{\alpha_{\text{SEIR}}} \delta_{\tau_i^*}(\tau) \tag{7}$$

i.e., a "spike" infectiousness profile where $\delta_i(\tau)$ is a delta function with mass at time t_i, and t_i is distributed according to the generation interval density $g(\tau) = A(\tau)/R_0$, such that a person's infectiousness is concentrated at a single moment (**Figure 1 A–C**).

In general, any definition of $a_i(\tau)$ such that $E_i[a_i(\tau)] = A(\tau)$ will yield the same mean-field population-level epidemic dynamics; however, different choices of $a_i(\tau)$ may have important stochastic consequences (**Figure 1 D–I**), which are the main topic of this article.

Figure 1: The individual infectiousness profile impacts stochastic epidemic trajectories. (A-C) Schematic diagrams of three individual infectiousness profile families that yield equivalent SEIR-type mean-field dynamics: the smooth (A), stepwise (B), and spike (C) profiles (Eqs. 4, 6, 7). (D-I) Deterministic epidemic trajectories (blue) are numerical solutions of the mean-field renewal equation model (Eq. 1), equivalent to a standard SEIR model with latent period $1/\gamma_{\text{SEIR}} = XX$, recovery period $1/\alpha_{\text{SEIR}} = XX$, and reproduction number $R_0 = 6$. For each individual infectiousness profile family, 1,000 stochastic simulations were generated (grey), with the cumulative infections (D-F) and daily new infection counts (G-I) depicted.

2.2 *Properties and components of the individual infectiousness profile*

Like $A(\tau)$, $a_i(\tau)$ is related to the individual reproduction number ν_i and the individual intrinsic generation interval distribution $\xi_i(\tau)$ (compare Eqs. 2–3):

$$\nu_i = \int_0^\infty a_i(\tau) d\tau \tag{8}$$

$$\xi_i(\tau) = \frac{a_i(\tau)}{\nu_i} \tag{9}$$

Re-arranging Eq. 9,

$$a_i(\tau) = \nu_i \xi_i(\tau) \tag{10}$$

which is a decomposition of $a_i(\tau)$ into infection intensity (ν_i) and timing ($\xi_i(\tau)$). This decomposition assumes that interpersonal contacts are uniform over time — or else that any non-uniformity is captured implicitly in ν_i and/or $\xi_i(\tau)$. A futher decomposition is possible by introducing an explicit contact process $c_i(t)$, defined with respect to the reference contact process $\bar{c}$ against which both R_0 and ν_i are defined [Dietz, 1993]. Let $c_i(t)$ be a nonnegative function describing person i's proportional deviations from $\bar{c}$ (so $c_i(t) \equiv 1$ represents the reference process), and let $b_i(\tau)$ be a density capturing the timing of person i's secondary infection attempts under $\bar{c}$. Then, we can define

$$a_i(t_i, \tau) = \nu_i b_i(\tau) c_i(t_i + \tau) \tag{11}$$

where t_i is person i's infection time and τ represents time since infection. For ν_i to retain its meaning, we require $E_t[c_i(t)] = 1$, that is, for a randomly chosen time t, the expected value of person i's contact modulator is 1. This is satisfied, for example, by a periodic function with average value 1 or by a stationary random process with mean 1. Another way of interpreting the contact modulator is that $\nu_i c_i(t)$ is the expected number of people that person i would infect if all their infectiousness were concentrated at time t (*i.e.*, if b_i were a delta function with mass at time t). The density $b_i(\tau)$ therefore reflects the "biological" component of a person's infectiousness timing (*i.e.*, the timing of secondary infections under a time-uniform reference contact process), and $c_i(t)$ captures the "contact-related" component of a person's infectiousness timing, while ν_i modulates the person's total infectiousness, which may be composed of both biological (*e.g.*, greater viral shedding) and contact-related (*e.g.*, persons with many social contacts) factors.

This extends back to the population-level model. If we assume that c_i is independent of ν_i and b_i, but allow ν_i and b_i to depend on one another (for example, we might expect a person with high overall infectiousness to also have a longer duration of infectiousness), then the population-level infectiousness profile is

$$A(t, \tau) = E_i[a_i(t, \tau)] = E_i[\nu_i b_i(\tau)] E_i[c_i(t + \tau)] \tag{12}$$

with the expectation taken over the population indexed by i. We define $C(t) = E_i[c_i(t)]$ to be the population-mean contact modulator at calendar time t. We define $B(\tau) = E_i\left[b_i(\tau)\frac{\nu_i}{R_0}\right]$, *i.e.*, the population-average biological infection timing density weighted by the person's infectiousness relative to the population mean. With these definitions, we obtain

$$A(t, \tau) = R_0 B(\tau) C(t + \tau) \tag{13}$$

Note that, like $c_i(t)$, $C(t)$ has time-average 1, since $E_t[C(t)] = E_t[E_i[c_i(t)]] = E_i[E_t[c_i(t)]] = 1$ (Fubini). If $C(t) \equiv 1$, we recover the original expression of $A(\tau) = R_0 B(\tau) = R_0 g(\tau)$, where $B(\tau)$ becomes the intrinsic generation interval distribution.

2.3 *A time-shifted Gamma model for biological infectiousness profiles with arbitrary punctuation*

Our central goal is to examine how the shape of the individual infectiousness profile $a_i(\tau)$ impacts epidemic dynamics, holding all else (including R_0 and $g(\tau)$) constant. The "stepwise" expression for $a_i(\tau)$ in Eq. 4 poses a difficulty: it introduces inherent overdispersion in the secondary infection distribution ($\nu_i =$

$\int_0^\infty a_i(\tau)d\tau \sim XX$), while for the smooth and spike profiles, ν_i is uniform across individuals ($\nu_i = R_0 \forall i$). This overdispersion affects stochastic epidemic dynamics, including the probability of epidemic establishment [Lloyd-Smith et al., 2005]. The impact of overdispersion on stochastic epidemic dynamics has been studied extensively elsewhere [Lloyd-Smith et al., 2005]. Here, we seek a family of functions that do not feature this inherent overdispersion, using the smooth and spike infectiousness profiles as book-ends.

Let us constrain the generation interval distribution $g(\tau)$ to be unimodal, and let us seek a family of functions $b_i(\tau)$ that are also unimodal with the narrowness of the function (variance of the distribution) modulated by a parameter $\psi \in [0,1]$, where $\psi \to 0$ yields a delta function (the spike case) and $\psi \to 1$ makes $b_i(\tau) \equiv g(\tau) \forall i$ (the smooth case). If we assume $g(\tau)$ is Gamma-distributed, which is a common choice for modeling generation interval distributions, the setup is straightforward. Specifically, let $g(\tau)$ be the Gamma(α, β) pdf, where $\beta = \alpha/\bar{g}$. Here, $\bar{g}$ is the mean generation time and α controls the shape of the generation interval distribution. A key property of the Gamma distribution is that the sum of two Gamma-distributed random variables with shape parameters α_1 and α_2 and common rate β is also Gamma-distributed with shape $\alpha_1 + \alpha_2$ and rate β. Let us assume now that the individual-level biological infectiousness profile $b_i(\tau)$ follows a shifted Gamma distribution preceded by a latent period l_i, which is also Gamma-distributed, *i.e.*:

$$b_i(\tau) = \begin{cases} 0 & \tau \leq l_i \\ f_\epsilon(\tau - l_i) & \tau > l_i \end{cases} \tag{14}$$

where $l_i \sim$ Gamma$((1-\psi)\alpha, \beta)$ and $f_\epsilon(\tau)$ is a Gamma$(\psi\alpha, \beta)$ pdf. The parameter $\psi \in [0,1]$ determines the punctuatedness of $b_i(\tau)$, with $\psi \to 0$ making $b_i(\tau)$ a delta function concentrated at $\tau = l_i$, and $\psi \to 1$ making $b_i(\tau) = g(\tau)$. As such, this provides a continuum between the "delta" and "smooth" cases from before (Eqs. 6-7). Due to the additive property of Gamma distributions with equal rate, these individual-level profiles are guaranteed to converge to the population-level $g(\tau) \sim$ Gamma(α, β). Furthermore, this convergence does not depend on the entire population sharing the same ψ-value; indeed, unique ψ_i may be chosen from any distribution with support on $[0,1]$ and the convergence holds (**Supplementary Methods**).

2.4 *Analytic approach*

To investigate the impact of the punctuatedness of $a_i(\tau)$ on epidemic dynamics, we turn to both analytic results and stochastic simultions. Stochastic simulations are generated by assuming a fixed population size (10,000) and initial number of infectious individuals (1). For each person i, a biological and contact infectiousness potential (b_i and c_i), and thus the full individual infectiousness profile (a_i), are generated *a priori*. Infection attempts are generated for each person by drawing a number of infections $X \sim$ Poisson(ν_i), where $\nu_i = \int_0^\infty a_i(\tau)d\tau$. These infections are then distributed according to $a_i(\tau)$ using Gamma-distributed draws. Infection attempts are paired with "recipients" uniform-randomly: if the recipient is susceptible, infection succeeds, while if the person has already been infected, the infection attempt fails. Full deatils are given in the **Supplementary Methods**. For illustration purposes, we focus on three sets of epidemiological parameters, roughly pertaining to influenza-like, SARS-CoV-2 omicron-like, and measles-like pathogens (**Table XX**).

We normally keep the contact process fixed at the reference value ($c_i(t) \equiv 1$). However, to examine the interaction between $b_i(\tau)$ and $c_i(t)$, we introduce a "periodic" and a "Pareto" contact model.

For the periodic contact model, we assume

$$c_i(t) = 1 - A\cos(\omega t) \quad \forall i \tag{15}$$

with amplitude $A \in [0,1]$ and angular frequency $\omega > 0$. We take $\omega = 2\pi\,\mathrm{day}^{-1}$ as the default, which models a contact process with a period of one day, reflecting higher numbers of contacts during the day and lower numbers at night. Setting $A = 0$ recovers the reference contact process $c_i(t) \equiv 1$. Note that this contact process also yields a periodic population-level contact modulator: $E_i[c_i(t)] = 1 - A\cos(\omega t)$. Here, there is no variation between individuals' contact processes, but there is variation over time.

For the Pareto contact model, we assume each person's contact modulator is a piecewise-constant process that switches to a new level at the points of a Poisson process with rate λ, *i.e.*, the sojourn at each level is exponentially distributed with mean $1/\lambda$. The levels that are switched to are independently drawn from a Lomax (Pareto Type II) distribution with shape α_P and scale σ,

$$f_{c^*}(c) = \frac{\alpha_P}{\sigma}\left(1 + \frac{c}{\sigma}\right)^{-(\alpha_P+1)}, \qquad c \geq 0 \tag{16}$$

with $\alpha_P > 1$. The Lomax distribution has support $[0,\infty)$ and mean $\sigma/(\alpha_P - 1)$, so requiring a mean of 1 (to be consistent with the reference contact process) pins down

$$\sigma = \alpha_P - 1 \tag{17}$$

leaving a one-parameter family indexed by the tail exponent α_P. Under this constraint, the variance of c^* collapses to $\mathrm{Var}[c^*] = \alpha_P/(\alpha_P - 2)$, which is finite for $\alpha_P > 2$ and diverges as $\alpha_P \searrow 2$; smaller α_P therefore corresponds to a heavier upper tail and rarer but more extreme "superspreader" contact levels, while larger α_P makes c^* progressively tighter around 1. Because each person's contact modulator is an iid sequence of Lomax levels held over exponential sojourns, the long-run time average satisfies $\lim_{T\to\infty} T^{-1}\int_0^T c_i(t)\,dt = E[c^*] = 1$ almost surely, so ν_i retains its interpretation as person i's expected number of secondary infections under the reference contact process. For this contact model, each person's infectiousness profile is unique, but the population-average infectiousness profile $C(t) = E_i[c_i(t)] = 1$, meaning the population-level, mean-field dynamics follow the reference contact process.

3 The impact of punctuated infectiousness profiles on uncontrolled epidemics

3.1 Punctuated individual infectiousness profiles generate later and more variably-timed epidemics.

Early-outbreak stochasticity can impact when an epidemic becomes established and begins its exponential growth phase [Morris et al., 2024]. This time shift durably impacts the epidemic's full trajectory. The effect of this early-outbreak stochasticity can be modeled as a random time shift applied to the deterministic epidemic trajectory [Morris et al., 2024]. Here, we demonstrate that the distribution of this time shift — both its mode and variance — depend on ψ.

Let contacts be uniform over time and across individuals ($c_i(t) \equiv 1 \, \forall \, i$), and let $b_i(\tau)$ follow the time-shifted Gamma model (Eq. 14). Drawing upon branching process theory, the cumulative incidence over time, $Z(t)$, during the exponential growth phase can be approximated as a deterministic exponential function with a random initial scaling factor W:

$$Z(t) \approx W \cdot e^{rt} \tag{18}$$

Here, r is the early-outbreak exponential growth rate and $E[W]$ is the initial number of infectious individuals, I_0. For clarity of the derivation, we use $E[W] = I_0 = 1$ throughout.

The random scaling factor W can, through a change of variables, be expressed as a random time shift:

$$Z(t) \approx e^{r(t+\varsigma)} \tag{19}$$

where

$$\varsigma = \frac{\log W - \log E[W]}{r} = \frac{\log W}{r} \tag{20}$$

when $E[W] = 1$.

Given this setup, it can be shown (**Supplementary Methods**) that

$$\mathrm{Var}[W] = \frac{\left(\frac{1+z}{1+2z}\right)^{\alpha} + \left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha} - 1}{1 - \left(\frac{1+z}{1+2z}\right)^{\alpha}} \tag{21}$$

where $z = R_0^{1/\alpha} - 1$. The punctuation parameter ψ enters the expression only once, in the middle term of the numerator; thus, $\mathrm{Var}[W]$ and ψ are monotonically related, where narrower individual infectiousness profiles ($\psi \to 0$) maximize the variance and broader profiles ($\psi \to 1$) minimize it.

We can also derive an approximate expression for the mode of the distribution of ς, $f_\varsigma(\tau)$. To approximate $f_\varsigma(\tau)$, we can use a two-moment version of the moment matching approach of Morris et al. [2024]. These moments are

$$E[W] = 1, \qquad E[W^2] = \frac{\left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha}}{1 - \left(\frac{1+z}{1+2z}\right)^{\alpha}} \tag{22}$$

where the first follows from the property $E[W] = I_0$ and the second follows from the variance calculation derived above, with $E[W^2] = \mathrm{Var}[W] + (E[W])^2 = \mathrm{Var}[W] + 1$. With these moments in hand it can be

shown (**Supplementary Methods**) that

$$\mathrm{Mode}[\varsigma] = \frac{1}{r}\log\left(\frac{2\left[1-\left(\frac{1+z}{1+2z}\right)^{\alpha}\right] - p\left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha}}{p\left[1-\left(\frac{1+z}{1+2z}\right)^{\alpha}\right]}\right) \tag{23}$$

provided the numerator within the logarithm is positive; when the numerator is zero or negative, no finite modal time shift exists and $\mathrm{Mode}[\varsigma] \to -\infty$. Again, ψ enters the expression only once, with narrow individual infectiousness profiles ($\psi \to 0$) minimizing the mode (making it more negative, *i.e.*, causing the typical epidemic to be shifted later) and broad individual infectiousness profiles ($\psi \to 1$) maximizing it (causing the typical epidemic to be shifted earlier).

Note that the ψ-dependence of both $\mathrm{Var}[W]$ and $\mathrm{Mode}[\varsigma]$ enter through the expression

$$\left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha} \tag{24}$$

This base is monotonically increasing in R_0 (**Supplementary Methods**), with $R_0 = 1$ implying $z = 0$, causing the ψ-dependence to disappear. Thus, increasing R_0 amplifies the ψ-dependency of $\mathrm{Var}[W]$ and $\mathrm{Mode}[\varsigma]$.

[A comment on intuition behind these results: narrower infectiousness profiles make it so that the offspring all inherit variation in the latent period(?), inflating the overall variance of the process. In a few epidemics, we have explosive spread early on, causing really fast take-offs; but often we also miss the early transmissions that trickle in causing lots of late take-offs. For the mode, we have a minimum of order statistics story, where the earliest offspring tends to come later the more punctuated the profile is.]

Stochastic simulations confirm the analytic results. The time to 100 infections varies with ψ, especially for pathogens with high R_0 (SARS-CoV-2 omicron, measles). For measles, by 28 days, only xx% of simulated epidemics reached 100 infections by day 28 for the spike case, while xx% reached the same threshold for the smooth case. Thus, punctuation of the individual infectiousness profile can substantially impact the timing of epidemic establishment, especially when R_0 is high.

3.2 Narrow individual infectiousness profiles yield greater uncertainty in epidemic growth rate estimates

Punctuated individual infectiousness profiles also make empirical estimates of the early-epidemic exponential growth rate r more uncertain (**Figure XX**).

3.3 Narrow biological infectiousness profiles interact with periodic contact patterns to generate overdispersion in secondary infections

When contacts are uniform in time, the "punctuatedness" of biological infectiousness (ψ) has no effect on the secondary infection distribution: every individual produces a Poisson(R_0)-distributed number of infections, and there is no overdispersion ($k \to \infty$) irrespective of ψ. However, if contacts vary over time, narrower biological profiles "see" more of this contact variation, generating overdispersion in the secondary infection distribution. We illustrate this using the periodic contact model, $c_i(t) = 1 - A\cos(\omega t + \Phi)$, for

which a closed-form expression for k can be obtained (**Supplementary Methods**):

$$k = \frac{2}{A^2}\left(1 + \frac{\omega^2}{\beta^2}\right)^{\psi\alpha} \tag{25}$$

where A is the amplitude of the contact oscillation and ω is its angular frequency. Here, the prefactor $2/A^2$ sets the overall overdispersion scale, and the bracketed factor quantifies the failure of the biological profile's temporal window to average out the contact oscillation, with $(1 + \omega^2/\beta^2)^{-\psi\alpha}$ being exactly the squared characteristic function of the Gamma$(\psi\alpha, \beta)$ density at the contact frequency ω. Since this factor is strictly increasing in ψ for any $\omega > 0$, narrower biological profiles always produce more overdispersion in the presence of periodic contact variation. The effect disappears in the spike limit only if $\omega \to 0$ or $A \to 0$ (no temporal contact variation), and grows arbitrarily large in the smooth limit if ω/β is large (contact oscillations much faster than the profile window, so the profile averages them out). Simulations extend this result to heavier-tailed, non-periodic contact heterogeneity, for which the same mechanism holds qualitatively (**Figure XX**).

4 The impact of punctuated individual infectiousness on detect-and-isolate interventions

The impact of a detect-and-isolate protocol can be measured in terms of its "test efficacy" (TE) [Middleton and Larremore, 2024]. The TE measures the amount of infectiousness that a detection protocol (possibly test-based, but this extends also to symptom-based isolation and other general isolation triggers) prevents.

When detection is unrelated to the timing of a person's infectiousness, the individual infectiousness profile cancels out, and the generation interval distribution is all that is neede to calculate TE. In the realistic scenario that detectability is anchored to the person's infectiousness somehow —*e.g.*, symptoms arise at a time related to peak infectiousness, or detectability can occur during some window that extends before and after a person's peak infectiousness – the shape of the individual infectiousness profile impacts TE. In general, when detection tends to occur before peak infectiousness, TE is higher for narrower infectiousness profiles; when detection usually happens after peak infectiousness, TE is lower for narrow infectiousness profiles (**Figure XX**).

TE, however, only reflects part of the story with individual infectiousness. Detect-and-isolate protocols have two additional impacts whose strength depends on the shape of the individual infectiousness profile: they re-shape the overall generation interval distribution by cutting off late transmissions (this happens moreso for smooth profiles than for spiky ones), and they can increase overdispersion in the secondary infection distribution (this happens moreso for smooth profiles). As a result, the naive simulations of epidemics using $R_{\text{eff}} = R_0(1 - \text{TE})$ fail to capture key elements of epidemics that result when the detect-and-isolate protocol is explicitly modeled and the individual infectiousness profile is accounted for.

We illustrate this using two specific detect-and-isolate protocols: one in which symptoms arise with some lag μ relative to peak infectiousness and variance σ^2, and one where there exists a detectability window around the peak infectiousness and people participate in regular testing.

Results for g, k, and TE for symptom-based screening.

Results for g, k, and TE for test-based screening.

Epidemic simulations illustrating differences.

5 The impact of individual infectiousness on gathering size restrictions

6 Identifiability and inference of the individual infectiousness profile's shape

While the concentration of the individual infectiousness profile can impact how an epidemic unfolds, we lack frameworks for estimating it. Many approaches exist for estimating the generation interval, but there has not been a systematic approach developed to assess how individual infectiousness varies and to specify where on the continuum we sit—i.e., how much of the variation in the generation interval distribution is attributable to variation between individuals, and how much to variation in infection timing within individuals.

We begin by assessing under which conditions we can hope to infer the shape of the individual infectiousness profile. We restrict to gamma-distributed timings, and consider cases where we assume that ψ is shared across individuals and where ψ comes from a beta distribution governed by hyperparameters α and β. Then, when ψ is identifiable, we introduce a hierarchical Bayesian inference strategy to infer the shape of the individual infectiousness profile given contact tracing data.

For this section: let's imagine that there's a single *psi* value shaed among everyone. We can use a Beta prior. Two things to consider: how do we estimate ψ given a list of infection times (assume we know them exactly)? And how do we estimate ψ given infection times and knowledge of $c_i(t)$? My hope is that this will emphasize that knowing the "negatives" – i.e., the contacts that "should" have led to transmission but did not – improves our inferential power substantially. Then, I want to think about how to achieve a given power in each scenario across R0, assuming some fixed generation interval distribution. That should be enough for now. We should probably include some amount of uncertainty in infection times for this base analysis, and we should post code for running the analysis given a line list of infection times and contacts. In the Discussion, we need to address the fact that there could be different ψ_i values across individuals, themselves drawn from a distribution, so we'd want some kind of hierarchical Bayesian model for inference here; also, we don't consider differences in the height (i.e. inherent biological differences in ν_i / b_i), which is another key thing to include in the inference.

7 Discussion

Uncertain early exponential growth rate calculations feed into estimates of R_0 — they are a critical ingredient for most R_0 estimation formulae.

The superspreading mechanism introduced here is novel, and implies that some amount of superspreading cannot be traced *a priori* to the individual: there is no such thing as identifying superspreaders, since the superspreader in this case may not have any more inherent infectiousness or a greater number of contacts; instead, they were just infected at the wrong time.

Accounting for the individual infectiousness profile is critical for anticipating the imapct of outbreak interventions. Simply calculating reductions in infectiousness is not enough: we get differences in overdispersion and the generation interval distribution that are sensitive to the punctuatedness of $a_i(\tau)$ and can lead to profoundly different epidemic dynamcis than a naive approach of simulating an equivalent epidemic without changing overdispersion or the generation interval distribution.

Estiamting the shape of the individual infectiousness profile can be challenging given the need for precise estimates of time of infection and a good balance of depth and breadth of sampling. Fortunately, identifiability of the shape of $a_i(\tau)$ is better in situations where it matters most, *i.e.*, when R_0 is high.

Limitations.

Concluding thoughts.

References

D. Breda, O. Diekmann, W. F. De Graaf, A. Pugliese, and R. Vermiglio. On the formulation of epidemic models (an appraisal of Kermack and McKendrick). *Journal of Biological Dynamics*, 6(sup2):103–117, September 2012. ISSN 1751-3758, 1751-3766. doi: 10.1080/17513758.2012.716454. URL http://www.tandfonline.com/doi/abs/10.1080/17513758.2012.716454.

David Champredon and Jonathan Dushoff. Intrinsic and realized generation intervals in infectious-disease transmission. *Proceedings of the Royal Society B: Biological Sciences*, 282(1821):20152026, December 2015. ISSN 0962-8452, 1471-2954. doi: 10.1098/rspb.2015.2026. URL https://royalsocietypublishing.org/doi/10.1098/rspb.2015.2026.

K. Dietz. The estimation of the basic reproduction number for infectious diseases. *Statistical Methods in Medical Research*, 2(1):23–41, March 1993. ISSN 0962-2802, 1477-0334. doi: 10.1177/096228029300200103. URL https://journals.sagepub.com/doi/10.1177/096228029300200103.

Ashish Goyal, Daniel B Reeves, E Fabian Cardozo-Ojeda, Joshua T Schiffer, and Bryan T Mayer. Viral load and contact heterogeneity predict SARS-CoV-2 transmission and super-spreading events. *eLife*, 10:e63537, February 2021. ISSN 2050-084X. doi: 10.7554/eLife.63537. URL https://elifesciences.org/articles/63537.

William Ogilvy Kermack and A. G. McKendrick. A contribution to the mathematical theory of epidemics. *Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character*, 115(772):700–721, August 1927. ISSN 0950-1207, 2053-9150. doi: 10.1098/rspa.1927.0118. URL https://royalsocietypublishing.org/rspa/article/115/772/700/2165/A-contribution-to-the-mathematical-theory-of.

Daniel B. Larremore, Bryan Wilder, Evan Lester, Soraya Shehata, James M. Burke, James A. Hay, Milind Tambe, Michael J. Mina, and Roy Parker. Test sensitivity is secondary to frequency and turnaround time for COVID-19 screening. *Science Advances*, 7(1):eabd5393, January 2021. ISSN 2375-2548. doi: 10.1126/sciadv.abd5393. URL https://www.science.org/doi/10.1126/sciadv.abd5393.

J. O. Lloyd-Smith, S. J. Schreiber, P. E. Kopp, and W. M. Getz. Superspreading and the effect of individual variation on disease emergence. *Nature*, 438(7066):355–359, November 2005. ISSN 0028-0836, 1476-4687. doi: 10.1038/nature04153. URL https://www.nature.com/articles/nature04153.

Casey Middleton and Daniel B. Larremore. Modeling the transmission mitigation impact of testing for infectious diseases. *Science Advances*, 10(24):eadk5108, June 2024. ISSN 2375-2548. doi: 10.1126/sciadv.adk5108. URL https://www.science.org/doi/10.1126/sciadv.adk5108.

Dylan Morris, John Maclean, and Andrew J. Black. Computation of random time-shift distributions for stochastic population models. *Journal of Mathematical Biology*, 89(3):33, September 2024. ISSN 0303-6812, 1432-1416. doi: 10.1007/s00285-024-02132-6. URL https://link.springer.com/10.1007/s00285-024-02132-6.

Åke Svensson. A note on generation times in epidemic models. *Mathematical Biosciences*, 208(1):300–311, July 2007. ISSN 00255564. doi: 10.1016/j.mbs.2006.10.010. URL https://linkinghub.elsevier.com/retrieve/pii/S0025556406002094.

J Wallinga and M Lipsitch. How generation intervals shape the relationship between growth rates and reproductive numbers. *Proceedings of the Royal Society B: Biological Sciences*, 274(1609):599–604, February 2007. ISSN 0962-8452, 1471-2954. doi: 10.1098/rspb.2006.3754. URL https://royalsocietypublishing.org/doi/10.1098/rspb.2006.3754.

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Supplementary Information

Supplementary Methods

Parameter structure

The punctuated Gamma model has four epidemiological parameters: the basic reproduction number R_0, the early-epidemic growth rate r, and the two parameters of the generation interval distribution, α (shape) and β (rate), which together determine the mean generation time $\bar{g} = \alpha/\beta$. These four parameters are not all free: the Euler–Lotka equation

$$1 = R_0 \int_{\tau=0}^{\infty} e^{-r\tau} \frac{\beta^\alpha}{\Gamma(\alpha)} \tau^{\alpha-1} e^{-\beta\tau} d\tau = R_0 \left(\frac{\beta}{\beta + r} \right)^\alpha \tag{S1}$$

provides one constraint among them, leaving three free parameters. We choose to work with $(R_0, \bar{g}, \alpha)$ as the free parameters, since each has a direct epidemiological interpretation:

- R_0: the basic reproduction number, measuring transmissibility;
- $\bar{g}$: the mean generation time;
- α: the shape parameter of the generation interval distribution, where larger α corresponds to a more concentrated (less variable) generation interval distribution for fixed $\bar{g}$.

Given $(R_0, \bar{g}, \alpha)$, the remaining parameters are fully determined. The rate parameter β follows immediately from the definition of the mean:

$$\beta = \frac{\alpha}{\bar{g}}, \tag{S2}$$

and the early-epidemic growth rate r follows from the Euler–Lotka equation:

$$r = \beta \left(R_0^{1/\alpha} - 1 \right) = \frac{\alpha}{\bar{g}} \left(R_0^{1/\alpha} - 1 \right). \tag{S3}$$

The punctuation parameter $\psi \in [0, 1]$ is the only remaining free parameter, and it is the sole quantity that is not identifiable from population-level epidemic data: since $A(\tau)$ is invariant across ψ by construction, the parameters $(R_0, \bar{g}, \alpha)$ can in principle be estimated from the observed epidemic trajectory independently of ψ. The consequences of varying ψ while holding $(R_0, \bar{g}, \alpha)$ fixed are therefore attributable purely to changes in the individual-level temporal structure of infectiousness, with no confounding from changes in the population-level transmission kernel.

Derivation of Var[W]

For a branching process $Z(t)$ conditional on non-extinction, $e^{-rt}Z(t) \to W$ almost surely as $t \to \infty$. Here, r is the early-epidemic exponential growth rate and W is a random variable that effectively scales the branching process' initial condition, capturing stochastic fluctuations in the early-epidemic process.

With a change of variables, we can translate W into a random initial time shift:

$$\varsigma = \frac{\log W - \log E[W]}{r} = \frac{\log W}{r} \tag{S4}$$

when $I_0 = E[W] = 1$. Our goal is to understand how $\mathrm{Var}[\varsigma]$ depends on the punctuation parameter ψ. Since ς and W are monotonically related, we can derive an expression for $\mathrm{Var}[W]$ and will interpret its implications for $\mathrm{Var}[\varsigma]$ at the end.

To start, we consider an index case i infected at time $t = 0$ who produces $\chi_i \sim \mathrm{Poisson}(R_0)$ offspring. Under the Gamma time-shift model, the offspring j from this index case are infected at times $\tau_j = l_i + \epsilon_j$, where l_i is person i's latent period and ϵ_j is the additional lag between l_i and person j's infection time, with

$$l_i \sim \mathrm{Gamma}((1-\psi)\alpha, \beta), \qquad \epsilon_j \sim \mathrm{Gamma}(\psi\alpha, \beta)$$

This ensures that the population-level generation interval distribution is

$$g(\tau) \sim \mathrm{Gamma}(\alpha, \beta)$$

We can define some useful quantities that will arise later in the derivation. First, from the Euler-Lotka equation under $g(\tau)$ [Wallinga and Lipsitch, 2007]:,

$$1 = R_0 \int_{\tau=0}^{\infty} e^{-r\tau} \frac{\beta^\alpha}{\Gamma(\alpha)} \tau^{\alpha-1} e^{-\beta\tau} d\tau$$

which simplifies to

$$1 = R_0 \Big(\frac{\beta}{\beta + r}\Big)^\alpha = R_0 \Big(\frac{1}{1+z}\Big)^\alpha$$

where $z = r/\beta = r\bar{g}/\alpha = R_0^{1/\alpha} - 1$, a dimensionless quantity that simplifies expressions and allows one to easily plug in different combinations of epidemiological parameters. Here, $\bar{g}$ is the mean of the generation interval distribution $g(\tau)$.

Next, we define

$$\rho_k = (E[e^{-kr\tau}])^{1/\alpha} = \frac{\beta}{\beta + kr} = \frac{1}{1+kz}$$

This allows us to re-write the Euler-Lotka equation:

$$1 = R_0 \rho_1^\alpha$$

Next, we introduce the distributional fixed point equation, which is a standard starting point for computing moments of W. This equation exploits the self-similarity of the branching process, in which an epidemic initiated by a single individual is, in distribution, a scaled and time-shifted copy of the whole epidemic. At large t, the epidemic seeded by the index case i consists of the downstream epidemics caused

by their offspring j, which have grown approximately to size $Z_j(t) = W_j e^{r(t-\tau_j)}$. That is,

$$Z(t) \approx \sum_{j=1}^{\chi_i} W_j e^{rt} e^{-r\tau_j}$$

Multiplying through by e^{-rt}:

$$W \stackrel{d}{=} \sum_{j=1}^{\chi_i} e^{-r\tau_j} W_j$$

We now use this expression to calculate $\mathrm{Var}[W]$. Since randomness enters through both W and τ, we first condition on τ_j using the law of total variance:

$$\mathrm{Var}[W] = E[\mathrm{Var}\Big[\sum_j e^{-r\tau_j} W_j | \tau_j\Big]] + \mathrm{Var}[E\Big[\sum_j e^{-r\tau_j} W_j | \tau_j\Big]]$$

$$= E\Big[\sum_j e^{-2r\tau_j} \cdot \mathrm{Var}[W]\Big] + \mathrm{Var}\Big[\sum_j e^{-r\tau_j} \cdot E[W]\Big]$$

$$= E\Big[\sum_j e^{-2r\tau_j}\Big] \cdot \mathrm{Var}[W] + (E[W])^2 \cdot \mathrm{Var}\Big[\sum_j e^{-r\tau_j}\Big]$$

Since $E[W] = I_0 = 1$, we have

$$\mathrm{Var}[W] = \mathrm{Var}[W] E\Big[\sum_j e^{-2r\tau_j}\Big] + \mathrm{Var}\Big[\sum_j e^{-r\tau_j}\Big]$$

Solving for $\mathrm{Var}[W]$,

$$\mathrm{Var}[W] = \frac{\mathrm{Var}[\sum_j e^{-r\tau_j}]}{1 - E[\sum_j e^{-2r\tau_j}]}$$

To simplify the denominator, note that

$$E\Big[\sum_{j=1}^{\chi_i} e^{-2r\tau_j}\Big] = E[\chi_i] E[e^{-2r\tau_j}] = R_0 E[e^{-2r\tau_j}] = R_0 \rho_2^\alpha$$

Thus, the variance expression simplifies to

$$\mathrm{Var}[W] = \frac{\mathrm{Var}[\sum_j e^{-r\tau_j}]}{1 - R_0 \rho_2^\alpha}$$

The denominator does not depend on ψ, so we proceed by examining the numerator:

$$\mathrm{Var}[\sum_j e^{-r\tau_j}] = E[(\sum_j e^{-r\tau_j})^2] - E[\sum_j e^{-r\tau_j}]^2 = E[(\sum_j e^{-r\tau_j})^2] - 1$$

Then,

$$E[(\sum_j e^{-r\tau_j})^2] = E[\sum_j e^{-2r\tau_j}] + E[\sum_{j \neq k} e^{-r(\tau_j + \tau_k)}]$$

$$= R_0 \rho_2^\alpha + E[\chi_i(\chi_i - 1)] E[e^{-r\tau_j} e^{-r\tau_k}]$$

$$= R_0\rho_2^\alpha + R_0^2 E[e^{-r\tau_j}e^{-r\tau_k}]$$

Here, the $\chi_i(\chi_i - 1)$ term enters because there are this many terms in the sum over $j \neq k$. Also,

$$E[\chi_i(\chi_i - 1)] = E[\chi_i^2] - E[\chi_i] = Var[\chi_i] + E[\chi_i]^2 - E[\chi_i] = R_0 + R_0^2 - R_0 = R_0^2$$

under the assumption that $\chi_i \sim \text{Poisson}(R_0)$.

Since $\tau_j = l_i + \epsilon_j$, and since l_i, ϵ_j, and ϵ_k are independent,

$$E[e^{-r\tau_j}e^{-r\tau_k}] = E[e^{-2rl_i}e^{-r\epsilon_j}e^{-r\epsilon_k}] = E[e^{-2rl_i}](E[e^{-r\epsilon}])^2 = \rho_2^{(1-\psi)\alpha} \cdot \rho_1^{2\psi\alpha}$$

Finally, we can assemble $\text{Var}[W]$:

$$\text{Var}[\sum_j e^{-r\tau_j}] = R_0\rho_2^\alpha + R_0^2\rho_2^{(1-\psi)\alpha}\rho_1^{2\psi\alpha} - 1$$

and thus, substituting in the z expressions for ρ_1 and ρ_2, we obtain

$$\text{Var}[W] = \frac{\left(\frac{1+z}{1+2z}\right)^\alpha + \left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha} - 1}{1 - \left(\frac{1+z}{1+2z}\right)^\alpha} \tag{S5}$$

which is Eq. 21.

Mode of the two-moment Gamma approximation for the time-shift distribution

Following a simplified version of the moment-matching approach of Morris et al. [2024], we approximate the distribution of W conditional on survival by a Gamma distribution matched to its first two moments. The unconditional moments of W are:

$$E[W] = 1 \tag{S6}$$

$$E[W^2] = \mathrm{Var}[W] + E[W]^2 = \frac{\left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha}}{1 - \left(\frac{1+z}{1+2z}\right)^{\alpha}} \tag{S7}$$

Conditioning on survival with probability p:

$$E[W \mid \mathrm{surv}] = \frac{E[W]}{p} = \frac{1}{p} \tag{S8}$$

$$E[W^2 \mid \mathrm{surv}] = \frac{E[W^2]}{p} = \frac{\left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha}}{p\left(1 - \left(\frac{1+z}{1+2z}\right)^{\alpha}\right)} \tag{S9}$$

where the division by p arises because

$$E[W^n] = E[W^n|\mathrm{surv}]P(\mathrm{surv}) + E[W^n|\mathrm{extinct}]P(\mathrm{extinct}) = E[W^n|\mathrm{surv}] \cdot p \tag{S10}$$

since $W = 0$ when the epidemic fails to establish. Thus, $E[W^n|\mathrm{surv}] = E[W^n]/p$.

Our goal is to approximate the distribution of W with a Gamma(k, λ) distribution, and specifically to estimate the mode of the distribution of $W|\mathrm{surv}$ with the mode of the moment-matched Gamma(k, λ) distribution. Note that the mode of a generic Gamma(k, λ) distribution is $(k-1)/\lambda$, and that the distribution has a mean of k/λ and a squared coefficient of variation of (CV^2) $1/k$. Thus, the parameters of the moment-matched Gamma distribution are

$$k = \frac{1}{\mathrm{CV}^2[W|\mathrm{surv}]} \tag{S11}$$

$$\lambda = \frac{k}{E[W|\mathrm{surv}]} = kp \tag{S12}$$

and the mode of the moment-matched distribution is

$$\mathrm{mode}[W|\mathrm{surv}] \approx \frac{k-1}{\lambda} = \frac{k-1}{kp} = \frac{1-1/k}{p} = \frac{1-\mathrm{CV}^2[W|\mathrm{surv}]}{p} \tag{S13}$$

The expression for CV^2 is

$$\mathrm{CV}^2(W|\mathrm{surv}) = \frac{\mathrm{Var}(W|\mathrm{surv})}{E[W|\mathrm{surv}]^2} = \frac{E[W^2|\mathrm{surv}] - E[W|\mathrm{surv}]^2}{E[W|\mathrm{surv}]^2} \tag{S14}$$

$$= \frac{p\left(\frac{(1+z)^2}{1+2z}\right)^{(1-\psi)\alpha}}{1 - \left(\frac{1+z}{1+2z}\right)^{\alpha}} - 1 \tag{S15}$$

This moment-matched approximation is valid when $k \geq 1$ (*i.e.*, $\mathrm{CV}^2[W|\mathrm{surv}] \leq 1$). When $k < 1$, the approximating density is monotonically decreasing on $(0, \infty)$ with no interior mode.

The time-shift $\varsigma = \log(W)/r$ is a monotone transformation of W, so its modal value is obtained by transforming the mode of the Gamma approximation. Therefore:

$$\mathrm{Mode}[\varsigma] = \frac{1}{r} \log\left(\frac{1-V}{p}\right) \tag{S16}$$

Substituting V from Eq. (S15) and expressing in terms of $z = R_0^{1/\alpha} - 1$, this yields Eq. (23).

Monotonicity in R_0

Differentiating Eq. (S22) with respect to R_0, holding α fixed, and using $\frac{d}{dR_0}(2R_0^{1/\alpha}) = (2/\alpha)R_0^{1/\alpha-1}$:

$$\frac{\partial(\alpha \log \sigma)}{\partial R_0} = \frac{2}{R_0} - \frac{2R_0^{1/\alpha-1}}{2R_0^{1/\alpha}-1} = \frac{2}{R_0} \cdot \frac{R_0^{1/\alpha}-1}{2R_0^{1/\alpha}-1} = \frac{2(\mu-1)}{R_0(2\mu-1)} \tag{S17}$$

This is strictly positive for all $R_0 > 1$, since $\mu = R_0^{1/\alpha} > 1$ makes both the numerator $\mu - 1$ and denominator $2\mu - 1$ positive. The derivative vanishes at $R_0 = 1$ ($\mu = 1$), confirming that at the epidemic threshold there is no ψ-dependence in either $\mathrm{Var}(W)$ or $\mathrm{Mode}[\varsigma]$.

Monotonicity in α

Differentiating Eq. (S22) with respect to α, holding R_0 fixed. Write $g(\alpha) = \alpha \log(2R_0^{1/\alpha} - 1)$, so that $\alpha \log \sigma = 2 \log R_0 - g(\alpha)$ and it suffices to show $\partial g / \partial \alpha > 0$. By the product rule, using $\frac{\partial}{\partial \alpha} R_0^{1/\alpha} = -R_0^{1/\alpha} \log R_0 / \alpha^2$:

$$\frac{\partial g}{\partial \alpha} = \log(2\mu - 1) - \frac{2\mu \log \mu}{2\mu - 1} \tag{S18}$$

where we have substituted $\mu = R_0^{1/\alpha}$ and $\log R_0 = \alpha \log \mu$. To establish the sign, define $h(\mu) = (2\mu - 1) \log(2\mu - 1) - 2\mu \log \mu$, so that the condition $\partial g / \partial \alpha > 0$ is equivalent to $h(\mu) > 0$. Differentiating:

$$h'(\mu) = 2 \log(2\mu - 1) + 2 - 2 \log \mu - 2 = 2 \log\left(2 - \frac{1}{\mu}\right) \tag{S19}$$

For $\mu > 1$, we have $2 - 1/\mu > 1$, so $h'(\mu) > 0$. Since $h(1) = 0$ and $h'(\mu) > 0$ for all $\mu > 1$, it follows that $h(\mu) > 0$ for all $\mu > 1$, giving $\partial g / \partial \alpha > 0$ and hence:

$$\frac{\partial(\alpha \log \sigma)}{\partial \alpha} < 0 \tag{S20}$$

So $\alpha \log \sigma$ is strictly decreasing in α: more concentrated generation intervals (larger α, holding R_0 fixed) reduce the impact of punctuation on both $\mathrm{Var}(W)$ and $\mathrm{Mode}[\varsigma]$.

The modulating role of $\alpha \log \sigma$

The ψ-dependent term in both $\mathrm{Var}(W)$ and $\mathrm{Mode}[\varsigma]$ is $\sigma^{(1-\psi)\alpha}$, which can be written as $e^{(1-\psi)\alpha \log \sigma}$. The quantity $\alpha \log \sigma$ therefore acts as the fundamental modulator of ψ-dependence: larger $\alpha \log \sigma$ means the base $\sigma > 1$ is raised to a larger effective power, amplifying the spread in $\mathrm{Var}(W)$ and $\mathrm{Mode}[\varsigma]$ between the spike ($\psi \to 0$) and smooth ($\psi \to 1$) limits. We can express this quantity exactly in terms of R_0 and α alone. Starting from $\sigma = \mu^2/(2\mu - 1)$ with $\mu = R_0^{1/\alpha}$:

$$\log \sigma = 2 \log \mu - \log(2\mu - 1) = \frac{2 \log R_0}{\alpha} - \log(2R_0^{1/\alpha} - 1) \tag{S21}$$

and multiplying through by α:

$$\alpha \log \sigma = 2 \log R_0 - \alpha \log(2R_0^{1/\alpha} - 1). \tag{S22}$$

We now show that this expression is strictly increasing in R_0 and strictly decreasing in α, confirming the qualitative picture that punctuation ($\psi < 1$) matters more when R_0 is large and the generation interval is dispersed (small α).

Summary

The two partial derivatives establish that the modulating quantity $\alpha \log \sigma$ — and hence the sensitivity of $\mathrm{Var}(W)$ and $\mathrm{Mode}[\varsigma]$ to ψ — is:

- **Increasing in** R_0: higher transmissibility amplifies the effect of punctuation, because more offspring more densely sample the individual infectiousness profile, inheriting more of its temporal structure.
- **Decreasing in** α: more concentrated generation intervals dampen the effect of punctuation, because there is less temporal variation across the generation interval for the shared latent period l_i to exploit.

Together, these results confirm that punctuated infectiousness ($\psi < 1$) has the greatest impact on epidemic timing variability when R_0 is large and the generation interval is dispersed — precisely the regime in which the individual infectiousness profile has the most temporal structure for offspring timing to inherit.

Overdispersion arising from periodic contact patterns

Here, we derive the Negative Binomial overdispersion parameter k [Lloyd-Smith et al., 2005] of the secondary infection distribution under the periodic contact model (Eq. 15), which we restate here in a stationary form with a common random phase $\Phi \sim \text{Uniform}[0, 2\pi)$:

$$c_i(t) = 1 - A\cos(\omega t + \Phi), \qquad A \in [0, 1]. \tag{S23}$$

Randomizing Φ makes $c_i(\cdot)$ stationary and is equivalent to assuming that the infection-time phase t_i mod $2\pi/\omega$ is uniform across individuals, which holds in good approximation during sustained exponential growth.

Lloyd–Smith bridge. The number of secondary infections from person i is $Z_i \sim \text{Poisson}(M_i)$ with $M_i = \int_0^\infty a_i(t_i, \tau)\, d\tau = \nu_i\, \tilde{c}_i$, where

$$\tilde{c}_i := \int_0^\infty b_i(\tau)\, c_i(t_i + \tau)\, d\tau. \tag{S24}$$

Matching the mean and variance of the mixed-Poisson offspring law to those of $\text{NegBin}(R_0, k)$ in the Lloyd-Smith parameterization gives $k = R_0^2/\text{Var}[M_i]$. With $\nu_i \equiv R_0$ held fixed (isolating the contribution of punctuation and contact variability), this reduces to

$$k = \frac{1}{\text{Var}[\tilde{c}_i]}. \tag{S25}$$

The rest of the derivation computes $\text{Var}[\tilde{c}_i]$.

Mean. Conditional on b_i, stationarity gives $E[c_i(t)] = 1$, hence

$$E[\tilde{c}_i \mid b_i] = \int_0^\infty b_i(\tau)\, d\tau = 1, \tag{S26}$$

using that b_i is a probability density. Therefore $E[\tilde{c}_i] = 1$ (and $E[M_i] = R_0$), and by the law of total variance $\text{Var}[\tilde{c}_i] = E\big[\text{Var}[\tilde{c}_i \mid b_i]\big]$.

Autocovariance. Averaging over the uniform random phase Φ,

$$\text{Cov}[c_i(t), c_i(t+s)] = A^2\, E_\Phi[\cos(\omega t + \Phi)\cos(\omega(t+s) + \Phi)] = \frac{A^2}{2}\cos(\omega s), \tag{S27}$$

so $\text{Var}[c_i(t)] = A^2/2$ and the spectrum of the contact process is concentrated at the single frequency ω.

Conditional variance of $\tilde{c}_i$. Substituting Eq. S27 into the double-integral form of the conditional variance,

$$\begin{aligned}
\text{Var}[\tilde{c}_i \mid b_i] &= \int_0^\infty \int_0^\infty b_i(\tau)\, b_i(\tau')\, \frac{A^2}{2}\cos\big(\omega(\tau - \tau')\big)\, d\tau\, d\tau' \\
&= \frac{A^2}{2}\,\text{Re}\left[\left(\int_0^\infty b_i(\tau)\, e^{i\omega\tau}\, d\tau\right)\left(\int_0^\infty b_i(\tau')\, e^{-i\omega\tau'}\, d\tau'\right)\right] \\
&= \frac{A^2}{2}\,\big|\hat{b}_i(\omega)\big|^2,
\end{aligned} \tag{S28}$$

where $\hat{b}_i(\omega) = \int_0^\infty b_i(\tau)\, e^{-i\omega\tau}\, d\tau$ is the Fourier transform (equivalently, the characteristic function at ω) of the biological profile.

Evaluating $|\hat{b}_i(\omega)|^2$. For the time-shifted Gamma model, $b_i(\tau) = f_\epsilon(\tau - l_i)$ with f_ϵ a Gamma$(\psi\alpha, \beta)$ density and random latent period l_i. A time shift multiplies the Fourier transform by a unit-modulus phase, so

$$\hat{b}_i(\omega) = e^{-i\omega l_i}\, \hat{f}_\epsilon(\omega), \qquad \left|\hat{b}_i(\omega)\right|^2 = \left|\hat{f}_\epsilon(\omega)\right|^2, \tag{S29}$$

with the l_i-dependence dropping out entirely. The characteristic function of Gamma$(\psi\alpha, \beta)$ is $\hat{f}_\epsilon(\omega) = (1 - i\omega/\beta)^{-\psi\alpha}$, giving

$$\left|\hat{f}_\epsilon(\omega)\right|^2 = \left(1 + \frac{\omega^2}{\beta^2}\right)^{-\psi\alpha}. \tag{S30}$$

Because Eq. S30 is deterministic, the outer expectation over b_i in the law of total variance is trivial:

$$\mathrm{Var}[\tilde{c}_i] = \frac{A^2}{2}\left(1 + \frac{\omega^2}{\beta^2}\right)^{-\psi\alpha}. \tag{S31}$$

Closed-form expression for k. Combining Eqs. S25 and S31,

$$\boxed{k = \frac{2}{A^2}\left(1 + \frac{\omega^2}{\beta^2}\right)^{\psi\alpha}.} \tag{S32}$$

The expression has two factors with transparent interpretations. The prefactor $2/A^2$ is the reciprocal variance of a uniformly-phased sinusoid with amplitude A, and it sets the overall scale: larger contact-oscillation amplitude $\Rightarrow$ smaller k $\Rightarrow$ more overdispersion. The bracketed factor $(1 + \omega^2/\beta^2)^{\psi\alpha}$ is the reciprocal of the squared characteristic function of b_i at the contact frequency ω, and it quantifies how much the biological profile's temporal window fails to average out the contact oscillation. This factor increases with ψ for any $\omega > 0$, so narrower profiles always produce more overdispersion in the presence of any periodic contact variation.

Limiting cases.

- **Spike profile, $\psi \to 0$:** the bracketed factor becomes 1, leaving $k = 2/A^2$. A spike profile samples c_i at a single instant, so the conditional variance of $\tilde{c}_i$ equals $\mathrm{Var}[c_i] = A^2/2$ regardless of ω or β.
- **Smooth profile, $\psi \to 1$:** $k = (2/A^2)(1 + \omega^2/\beta^2)^\alpha$, strictly larger than the spike value whenever $\omega > 0$.
- **Slow oscillation, $\omega/\beta \to 0$:** the contact process looks locally constant over the infection window, so $\tilde{c}_i \approx c_i(t_i)$ for any ψ, and $k \to 2/A^2$ independent of ψ.
- **Fast oscillation, $\omega/\beta \to \infty$:** the contact process averages out over any infection window with $\psi > 0$, driving $\mathrm{Var}[\tilde{c}_i] \to 0$ and $k \to \infty$ (Poisson).
- **No contact variation, $A \to 0$:** $k \to \infty$ trivially.

These limits are consistent with the intuition that overdispersion arises from incomplete averaging of contact variation by the biological profile window, and that the degree of averaging is controlled by both $\psi\alpha$ (profile width) and ω/β (ratio of profile timescale to contact timescale).

Supplementary Figures

Supplementary Tables